Data-Driven Frequency Bands Selection in EEG-based Brain-Computer Interface

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Abstract— **In this paper, we propose a novel method of frequency bands selection based on the analysis of a channel-frequency map, which we call 'channel-frequency map'. The spatial filtering, feature extraction, and classification processes are operated in each frequency band in parallel. We determine a class label for an input EEG based on the outputs from the multi-streams with a two-step decision strategy at the end. From our experiments on a public dataset of BCI Competition IV (2008) II-a that includes four motor imagery tasks from 9 subjects, the proposed algorithm outperformed the Common Spatial Pattern (CSP) algorithm and a filter bank CSP algorithm on average in terms of a session-to-session transfer rate using one session for training and the other session for test. A considerable increase of classification accuracy has been achieved for certain subjects. We also would like to note that the proposed data-driven frequency bands selection method is applicable to other single-trial EEG classification that is based on modulations of brain rhythms.**

Keywords - Brain-Computer Interfaces; frequency bands selection; motor imagery classification; electroencephalography; event-related (de)synchronization (ERD/ERS); machine learning

I. INTRODUCTION

The Brain-Computer Interfaces (BCIs) enable to establish a direct communication pathway between human intentions and external electronic devices via the translation of electrical brain signals into a user's commands without using muscles activity or peripheral nerves. Due to their great potential to medical and industrial applications, they have been considered as an emerging technology and have been of great interest to many research groups [1, 2, 3].

However, high complexity of a human brain and low Signal-to-Noise Ratio (SNR) of EEG signals prevent the BCI systems from decoding every human mental states or intents. Among various mental tasks, increasing attention has been devoted to the analysis of EEG signals evoked by Motor Imageries (MIs) due to its operational characteristics, *i.e.*, asynchronous and continuous. There is a well-known neuro-physiological phenomenon during MIs. There occurs suppression or enhancement of the power in the motor and somatosensory cortex due to the loss of synchrony in a particular frequency range, which are called Event-Related De-synchronization (ERD) and Event-Related Synchronization (ERS), respectively [1].

Brain rhythms related to the imagination of movement are known to broadly be concentrated in the frequency bands of μ (12-16*Hz*) and β (18-24*Hz*). The frequency bands, however, are highly variable over different MIs as well as inter-subjects. In terms of designing a MI-based BCI system, the selection of proper frequency bands in which we filter the EEG data before extracting features are one of the most challenging problems in BCIs. In general, a frequency range of importance in discrimination of EEG signals is identified by a visual analysis on the data transformed into a frequency domain requiring a time-consuming process for each set of mental tasks to be classified and for each subject.

To our knowledge, the frequency bands on which the process in a BCI system operates is either selected manually or unspecifically set to a broad band. Only a few papers in the literature have partially addressed the selection of frequency bands. Fazli focused on decoding EEG data for subject-independent BCI by constructing a library of subject-specific spatio-temporal filters from a large database of EEG recordings of 83 BCI users and deriving a subject independent classifier [4]. Dalponte *et al.* proposed a method of selecting time and frequency intervals for effective feature extraction in EEG signals for BCI applications by searching a huge space of quantized frequency and time intervals [5]. In order to address the problem of selecting the subject-specific frequency bands for the CSP algorithm, Ang *et al.*, proposed to dissect a broad frequency range of interest into small non-overlapping filter banks and applied CSP algorithm for each band, which they called as the Filter Bank Common Spatial Pattern (FBCSP) [6]. Later, Chin *et al.* extended it for multi-class MIs classification [7].

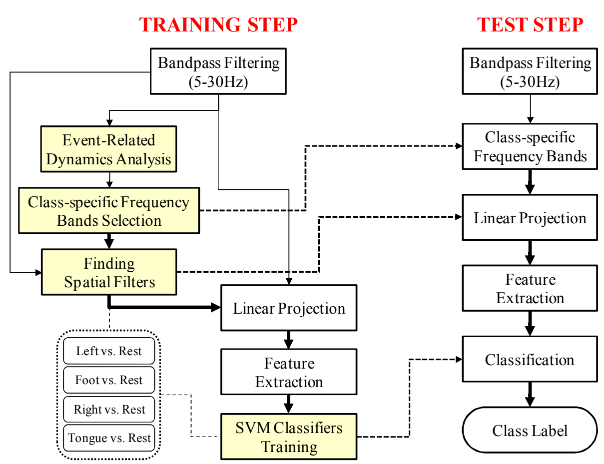

Figure 1. Overview of the proposed method.

In this paper, we propose a novel method of selecting subject- and class-specific frequency bands in the

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classification of MIs based on the time-frequency map analysis using Event-Related Spectral Perturbation (ERSP) [8]. In each of the selected frequency bands the classification process is operated individually. An overview of the proposed MIs classification system is illustrated in Fig. 1, where the thick solid arrows denote multi-streams operated in parallel.

The paper is organized as follows. Section II describes the mental tasks of our interest and preprocessing methods. In Section III, we propose a method of data-driven frequency bands selection, feature extraction, and classification. Experimental results are then presented in Section IV and conclusions are followed in Section V.

II. Dataset and Preprocessing

A. Mental Tasks and Experimental Paradigm

We use the dataset IIa of BCI Competition IV (2008) provided by the BCI research group at Graz University [9], which contains EEG signals recorded from 9 subjects performing four different MI tasks, *i.e.*, left-hand, right-hand, foot, and tongue, comprised of two sessions conducted on different days. Each session includes 6 runs separated by short breaks, and a run is further composed of 48 trials; there are 12 trials per MI task and 288 trials in total per session.

The EEG data were acquired using 22 AG/AgCl electrodes whose montage is presented in Fig. 2 (left). The signals were sampled at 250*Hz* and bandpass-filtered between 0.5*Hz* and 100*Hz*. An additional 50*Hz* notch filter was also applied to suppress line noise. The timing scheme of the experimental paradigm is depicted in Fig. 2 (right). The subjects were asked to carry out the MIs task corresponding to the cue in the form of an arrow pointing either to the left (left hand, class 1), right (right hand, class 2), down (foot, class 3) or up (tongue, class 4).

B. Data Preprocessing

The EEG signals were bandpass-filtered between 5*Hz* and 30*Hz* covering μ (8-13*Hz*) and β (14-30*Hz*) rhythms that are known as related to MIs. We then applied small Laplacian derivation calculated by subtracting four surrounding channels with weights equal to the central one, in order to remove artifacts and noise.

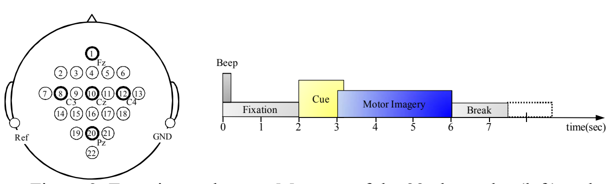

Figure 2. Experimental setup: Montage of the 22 electrodes (left) and timing scheme of the paradigm (right) [9].

III. Proposed Method

A. Mental Tasks and Experimental Paradigm

In this paper, we utilize an Event-Related Spectral Perturbation (ERSP), a generalization of the ERD/ERS attended during MI, to measure the event-related brain dynamics for the analysis of time-locked EEG spectrum induced by MI. An ERSP measures dynamic changes of EEG signals evoked by an experimental stimulus such as MI as a function of time in the broadband frequency spectrum [8]. The spectral changes involved in MIs can be more than one frequency bands depending on the subjects and the stimulus. This motivates us to analyze a full-spectrum ERSP and find informative frequency bands on brain dynamics rather than considering the predefined and fixed narrow-band ERD/ERS. Refer to [8] for the computation and detailed explanation of an ERSP.

B. Channel-Frequency Map (CFM)

The ERSP measures the brain dynamics of each channel independently. However, due to the nature of EEG measurement, neighboring electrodes may have common sources and spectral attributes that contribute to the measured EEG signals. We combine the multi-channel distributed ERSPs into a matrix by concatenating the time-domain averaged out ERSP of each channel as follows

$$CFM_i(C,F) = \left[X_i^1,\ldots,X_i^c,\cdots,X_i^C\right] \tag{1}$$

$$X_i^c = \frac{1}{T}\sum_{t=1}^{T} E_i^c(f,t) \tag{2}$$

where $E_i^c(f,t)$ represents an averaged ERSP at the frequency f and the time point t of c-th channel during i class MI, C and F denote, respectively, the number of channels and frequency components of interest. We call the matrix represented by Eq. (1) a '*Channel-Frequency Map* (*CFM*)'. Fig. 3 shows examples of the CFMs for subject 1 and subject 3. The X_i^c in Eq. (2), a time-domain averaged ERSP of the c-th channel, represents the mean amplitude of the ERD/ERS in an individual frequency component for the c-th channel.

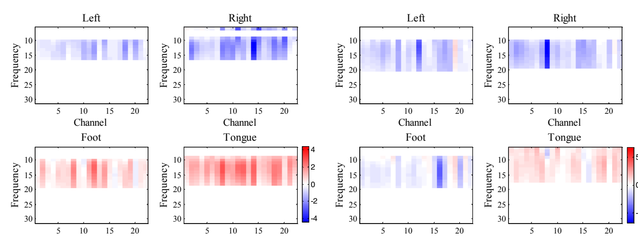

Figure 3. Examples of the Channel-Frequency Map (CFM) for subject 1 (left) and subject 3 (right).

We then compute a weight of the specific frequency component f' by normalizing the sum of the squared CFM_i over the channels with the sum of the squared CFM_i over both channels and frequencies as follows

$$w_i(f') = \frac{\sqrt{\sum_c CFM_i(c,f')^2}}{\sqrt{\sum_f \sum_c CFM_i(c,f)^2}} \tag{3}$$

Here $w_i(f')$ can be considered as an intensity of the ERD/ERS for the frequency component f'. If an ERD/ERS weight is larger than a threshold we believe that the corresponding frequency component is informative for i-th class and include it to the set of the selected frequency components:

$$SF_i = \bigcup_f \left(w_i(f) > \delta_i^f \right) \quad \forall f \in \{1,2,\ldots,F\} \tag{4}$$

where δ_i^f denote a threshold, which in our experiments we set as the mean of the weights.

C. *Feature Extraction*

For each of the selected frequency bands we build an independent classification stream as shown in Fig. 4. In each stream, we first apply band-pass filter over the range of the corresponding frequency band for all channels except the EOG. We consider the sample points of the interval between 0.5s and 2.5s after onset of the stimulus. We then apply the CSP algorithm [10] employing a one-versus-the-rest strategy to find the optimal spatial filters. The number of spatial filters is set to 6 empirically. Once the set of spatial filters has been determined we apply them to the training and test sets. We extract feature vectors of the logarithm of the variances on the spatially and spectrally filtered output signals. The feature vectors are then transformed by a multiple discriminant analysis to increase the discriminability among classes.

D. *Designing a Classifier*

A Support Vector Machine (SVM), which has proved to perform strongly in a number of real-world problems including BCI [11], is trained with the transformed data. Same as the CSP algorithm, we train the SVM with a one-against-all approach.

Since we have multiple streams, each of which produces an independent class label for the given EEG data, it needs to make a decision based on the resulting labels. We use a two-step decision strategy. The first step is a voting strategy as follows

$$\hat{v} = \arg\max_v V(v) \tag{5}$$

where $V(v)$ represents the number of classifiers that output the class label v for the given EEG data. If there is a bin that receives the largest votes from the SVMs, we then classify the input EEG into the corresponding label without proceeding to the second step. Or not, as a second step, we consider the confidence of each SVM. That is, if more than two classes have the same maximum votes, we choose the class label to which a distance from the given feature is the largest:

$$\hat{i} = \arg\max_i D(i) \quad \text{iff } |\hat{v}| \geq 2, \text{ where } i \in \hat{v} \tag{6}$$

where $D(i)$ denotes a maximal distance between the feature vector and the hyperplane of a classifier that output a class label $\hat{v}$ and $|v|$ is a number of elements in the set v. This strategy is from the rationale that the larger the distance from the hyperplane the more confident belonging to the class.

Fig. 4 schematizes the operational flow of the proposed method for four MIs classification, where the subscripts l, r, f, t, respectively, denote the number of frequency bands for the classes of left-hand, right-hand, foot, and tongue. Since we choose frequency bands of interest for each class individually, there exist $l+r+f+t$ independent streams. At the end of the system, we combine the outputs from the multi-stream SVM classifiers as depicted in Eq. (5) and Eq. (6) to decide a class label for an input EEG.

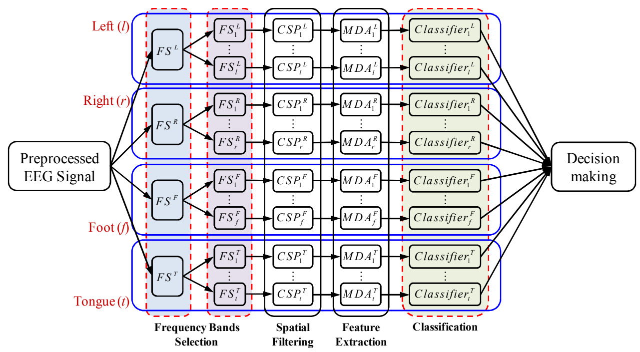

Figure 4. Flowchart of the proposed method for multi-class MI classification. The subscripts l, r, f, t, respectively, denote the number of frequency bands for the classes of left-hand, right-hand, foot, and tongue MI. FS (Frequency Selection), CSP (Common Spatial Pattern), MDA (Multiple Discriminant Analysis).

IV. RESUTLS AND ANALYSIS

A. *Competing Models*

In order to show the effectiveness of the proposed method, we build three competing models: BaseLine (BL), Chin *et al.*'s Filter Bank (FB) [7], and the proposed frequency selection. While the BL model deals with EEG signals bandpass-filtered between 5*Hz* and 30*Hz* covering μ (12-16*Hz*) and β (18-24*Hz*) rhythms, we should consider multiple bands for the other two models. In the FB and the proposed models, the ensuing operations are applied for each band individually, but we produce one class label by considering the outputs from multiple classifiers as explained in Section IV-D.

The CSP algorithm is applied to the preprocessed and bandpass-filtered EEG data in order to find optimal spatial filters. In our experiments, the number of spatial filters was set to 6 empirically, and applied equally to all the competing models. Feature extraction based on a multiple linear discriminant analysis and a classification with SVMs is followed. We used a nonlinear SVM with a Gaussian radial basis function. In the FB model, we set the size of a filter bank 4*Hz* following Chin *et al.*'s work [7].

B. *Session-to-Session Transfer*

The goal of the BCI Competition IV on the dataset II-a was to evaluate classification algorithms based on a session-to-session transfer rate, using session 1 dataset for training and session 2 dataset for evaluation. In Table 2, we summarize the performances of three competing methods. The proposed method outperformed the other methods for all subjects except for subject 2 when we used the EEG data of session 1 for training and the EEG data of session 2 for evaluation, and vice versa. We also compare the best performances obtained by the proposed method with the performances of the three best competitors and other methods recently published in Gouy-Pailler *et al.*'s work [12]. For a fair comparison with the other methods we employ the criterion of the kappa score, used in the competition. Table 3 summarizes the results. We can see that the proposed frequency bands selection method proved the best for four out of nine subjects and presented the second highest kappa score on average among 7 different methods. Here we should note that the first and third best competitors applied bandpass-filtering and linear regression, respectively, to remove the artifacts in the dataset.

TABLE I. KAPPA SCORES OF THE BEST PERFORMANCE OF THE PROPOSED METHOD AS WELL AS THREE BEST COMPETITORS IN BCI COMPETITION IV (2008) AND THE METHODS OF JAD (JOINT APPROXIMATE DIAGONALIZATION), CSP (COMMON SPATIAL PATTERN), AND MSJAD (MULTI-SEGMENT JAD) PRESENTED IN GOUY-PAILLER ET AL.'S WORK [12].

	Proposed	Three best competitors[1]			Gouy-Pailler *et al.* [12]		
		1st	2nd	3rd	JAD	CSP	MSJAD
Subject 1	**0.71**	0.68	0.69	0.38	0.65	0.52	0.66
Subject 2	0.31	0.42	0.34	0.18	0.40	0.39	**0.42**
Subject 3	0.75	0.75	0.71	0.48	**0.77**	0.67	**0.77**
Subject 4	0.47	0.48	0.44	0.33	0.50	0.50	**0.51**
Subject 5	0.19	0.40	0.16	0.07	0.44	0.49	**0.50**
Subject 6	0.20	**0.27**	0.21	0.14	0.19	0.18	0.21
Subject 7	**0.78**	0.77	0.66	0.29	0.25	0.26	0.30
Subject 8	**0.77**	0.75	0.73	0.49	0.72	0.57	0.69
Subject 9	**0.73**	0.61	0.69	0.44	0.50	0.40	0.46
Mean	0.55	0.57	0.51	0.31	0.49	0.44	0.50

V. CONCLUSIONS

In designing a MI-based BCI system, it requires a time-consuming process of frequency bands selection for each set of mental tasks to be classified and for each subject based on visual inspection. The previous approaches largely selected frequency bands manually or unspecifically set to a broad band. In this paper, we propose a data-driven method of subject- and class-specific frequency bands selection for multi-class MIs classification based on the time-frequency map analysis using Event-Related Spectral Perturbation (ERSP) [8]. In each of the selected frequency bands the classification process is operated individually in parallel and a label for an input EEG is determined by considering the outputs from multiple classifiers together at the end.

From our experiments on a public dataset of BCI Competition IV II-a of four motor imageries: left-hand, right-hand, foot, and tongue, the proposed method outperformed a baseline model that considered a wideband (between 5*Hz* and 30*Hz*) EEG and a Filter Bank (FB)-based model that followed Chin *et al.*'s approach [7]. Apart from the higher accuracy in classification, the proposed method has an advantage of eliminating a time-consuming calibration process of determining frequency bands for each subject. We also would like to remark that the proposed channel-frequency map is not limited to the motor imagery classification, but also applicable to other EEG-based brain computer interface, such as P300-based word speller.

[1] The results as well as the methods produced them were detailed on the website of BCI Competition IV (2008) (http://www.bbci.de/competition/iv/results/index.html)

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